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Creating Competitive Advantage

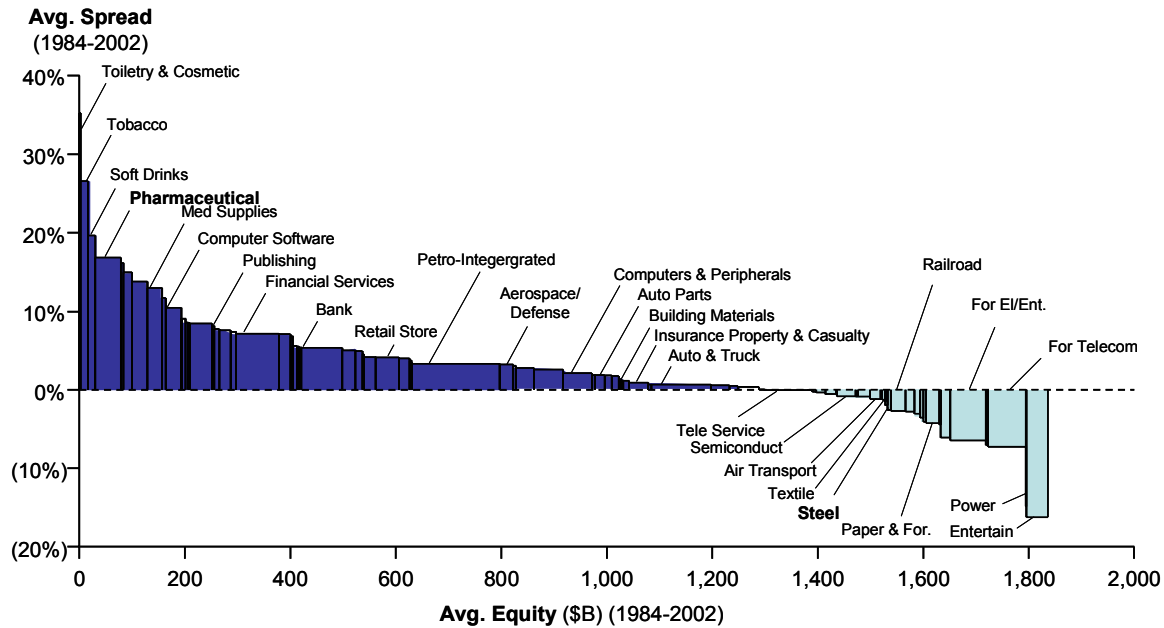
Some companies generate far greater profits than others. The pharmaceutical maker Schering-Plough produced an economic profit of more than \$10 billion during the period 1984-2002. That is, the accounting profit it generated exceeded its cost of equity capital by that amount. Over the same period, U.S. Steel produced an economic loss of nearly \$500 million; its cost of capital exceeded its accounting profit by a wide margin. Such large differences in economic performance are commonplace. Understanding their roots is crucial for strategists.

Differences in industry structure shed some light on such differences in performance.¹ To a certain extent, Schering-Plough has generated more economic profit than U.S. Steel because the pharmaceutical industry is structurally more attractive than the steel industry. Rivalry in the pharmaceutical market is muted by factors such as patent protection, product differentiation, and expanding demand; in contrast, rivalry in the steel industry is fierce—fueled by excess capacity, limited differences across products, and slow growth. Many pharmaceutical users hesitate to switch among products or brands, while steel customers are usually willing to switch among producers to get a better price. Many pharmaceuticals are made from commodities with little labor input, while unions exercise such power in the steel industry that labor costs often account for a quarter of total revenue. Such contrasts in industry-level competitive forces are one reason that the profit levels of firms in different industries differ. **Figure 1** shows, for each of many industries, the spread between the industry's return on equity and its cost of equity (the vertical axis) and the average equity in the industry (the horizontal axis) for the period 1984-2002. Reflecting differences in industry-level competitive forces, the pharmaceutical industry has been among the greatest generators of economic profit, while the steel industry as a whole has produced losses. The typical pharmaceutical maker is far more profitable than the typical steel producer.²

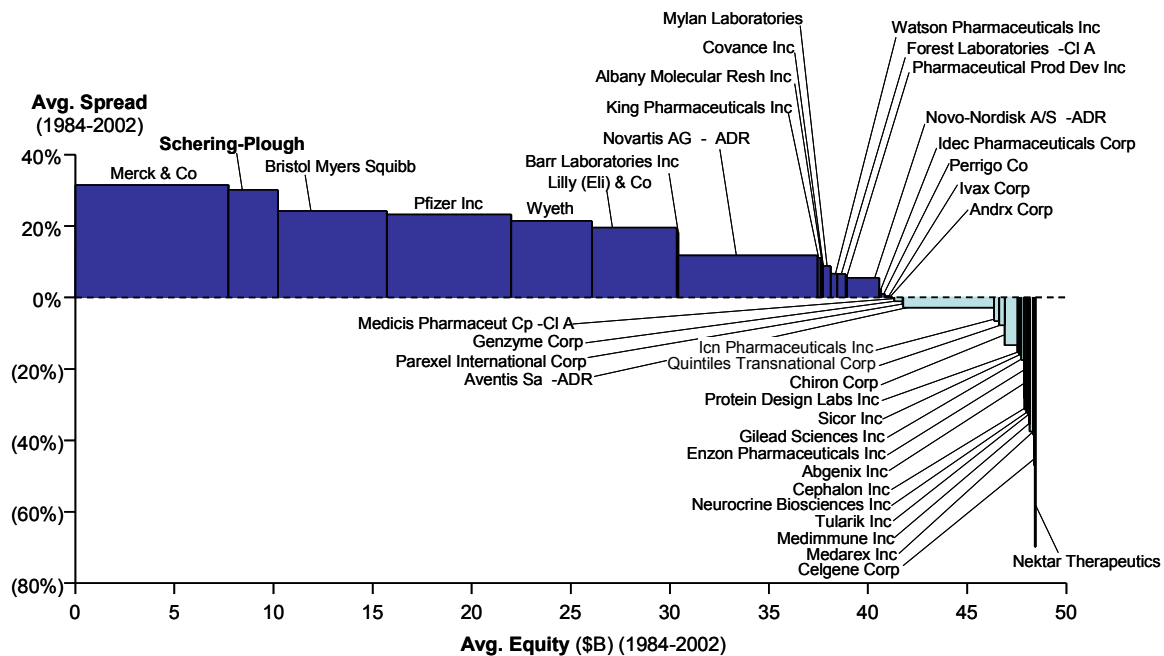
Schering-Plough, however, is not a “typical pharmaceutical maker,” nor is U.S. Steel a “typical steel producer.” As **Figures 2a** and **2b** illustrate, industry averages can mask large differences in economic profit within industries. Schering-Plough was far more effective at producing economic profits than were many drug makers during the 1984-2002 period, while U.S. Steel performed far worse than many other steel producers. Indeed, recent research indicates that intra-industry differences in profitability like those shown in **Figures 2a** and **2b** may be larger than differences across industries such as those in **Figure 1**.³ Industry-level effects appear to account for 10-20% of the variation in business profitability while stable within-industry effects account for 30-45%. (Most of the remainder can be assigned to effects that fluctuate from year to year.)

Professors Pankaj Ghemawat and Jan W. Rivkin prepared this note as the basis for class discussion. It is based in part on earlier notes by Pankaj Ghemawat, Tarun Khanna, and Anita McGahan.

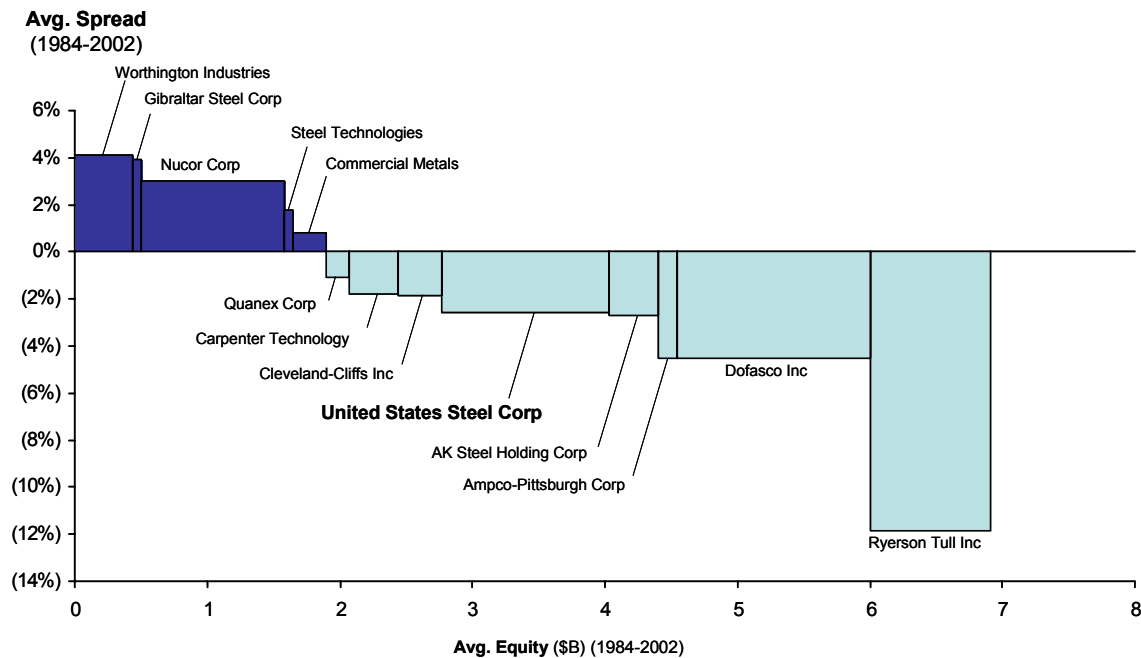
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Figure 1 Economic Profits of U.S. Industry Groups, 1984-2002

Source: Compustat, Value Line, Marakon Associates analysis

Figure 2a Economic Profits in the Pharmaceutical Industry, 1984-2002

Source: Compustat, Value Line, Marakon Associates analysis

Figure 2b Economic Profits in the Steel Industry, 1984-2002

Source: Compustat, Value Line, Marakon Associates analysis

In light of this, strategists need a systematic way to understand and analyze within-industry differences in performance. Toward that end, this note uses the notion of *competitive advantage*. A firm is said to have a competitive advantage over its rivals if it has driven a wide wedge between the willingness to pay it generates among buyers and the costs it incurs—indeed, a wider wedge than its competitors have achieved.⁴ A firm with a competitive advantage is positioned to earn superior profits within its industry. In examining the logic of how firms create competitive advantage, this note emphasizes two themes. First, to create an advantage, a firm must configure itself to do something unique and valuable. The firm must ensure that, were it to disappear, someone in its network of suppliers, customers, and complementors would miss it and no one could replace it perfectly.⁵ The first section of the note uses the concept of “added value” to make this point more precisely. Second, competitive advantage usually comes from the full range of a firm’s activities—from production to finance, from marketing to logistics—acting in harmony. The essence of creating advantage is finding an *integrated set* of choices that distinguishes a firm from its rivals. The second section of the note shows how managers can analyze the full range of activities to understand the sources of competitive advantage.

As a preface to the main discussion, it is important to address a few possible misconceptions.

Creating vs. sustaining competitive advantage. The note separates the challenge of creating competitive advantage at a point in time from the problem of sustaining advantage over time. In reality, the two issues are married: the choices that establish a firm’s advantage also influence whether the advantage can be sustained. For instance, in launching its personal financial software “Quicken,” Intuit chose to offer customers outstanding post-sale assistance over the telephone. Customers valued the help from trained operators, and customer service became a tool for creating competitive advantage. Moreover, customer service helped Intuit sustain its advantage over rivals such as Microsoft. Competitors found it hard to match Intuit’s service operations and its reputation

for excellent support. In addition, Intuit used information from its service operations to generate a stream of ideas for improving its product.⁶

Despite the connections between creating and sustaining advantage, we find it important to discuss the two processes separately. Each is so complicated that it would be unwieldy to deal with both at once.

Links to industry analysis. Within-industry differences in performance are often larger than differences across industries, but it would be wrong to conclude that industry analysis is unimportant. Industry analysis is crucial to creating competitive advantage for several reasons.

First, companies that generate competitive advantages typically do so by devising strategies that neutralize the unattractive features of their industries and exploit the attractive features.

Second, industry conditions appear to have a large influence on whether competitive advantages are even possible.⁷ In some industries (e.g., computer leasing), conditions “strait-jacket” firms and leave them little room to establish a superior wedge between willingness to pay and costs. In other industries (e.g., prepackaged software), conditions permit the most effective firms to enjoy large advantages over the least.

Finally, market leaders often face a tension between managing industry structure and pursuing an advantage within that structure. When deciding whether to build a new aluminum smelter, for instance, Alcoa must consider the impact of the new capacity on industry supply-demand conditions, not just its effect on Alcoa’s competitive advantage. This is true not only because Alcoa is a large player in the business, but also because Alcoa is closely tracked by its rivals.

Analysis and creativity. This note takes an analytical approach to competitive advantage. In actuality, many of the greatest advantages come not from analysis, but from entrepreneurial insight and trial-and-error. The cold, hard analysis described here is not intended to deny the importance of insight and trial-and-error. Rather, it aims to guide entrepreneurial creativity and to set a battery of tests for new business ideas.

The Logic of Value Creation and Distribution

The first and foremost test in this battery concerns “added value,” a concept developed by Adam Brandenburger, Barry Nalebuff, and Harborne Stuart.⁸ To introduce the concept, we use the example of the portal crane business of Harnischfeger Industries.⁹ We then link added value to competitive advantage.

Harnischfeger, based in Milwaukee, Wisconsin, manufactured equipment for industrial customers. Its material handling equipment division served a range of customers, including forest products companies such as International Paper. In the late 1970s, Harnischfeger began to offer these customers a new product: portal cranes. Portal cranes were designed to lift entire tree-length logs off of railcars and trucks and to hoist them around woodyards. The cranes were a significant improvement over the giant forklifts that they replaced.

In fact, it was possible to calculate the customer benefits reasonably precisely. Each crane replaced a fleet of forklifts which cost roughly \$1.0 million. A crane was also less expensive to operate than a forklift fleet; it required less labor, fuel, and maintenance, for instance. Altogether over its lifespan, each crane generated a net present value of \$6.5 million of savings in operating costs. It cost Harnischfeger only \$2.5 million to produce and install each crane. Thus a large gap

existed between the customer benefits associated with a crane (\$1.0 million + \$6.5 million) and Harnischfeger's costs (\$2.5 million). Despite this gap, Harnischfeger was making little profit on its sales of portal cranes by the late 1980s. What happened?

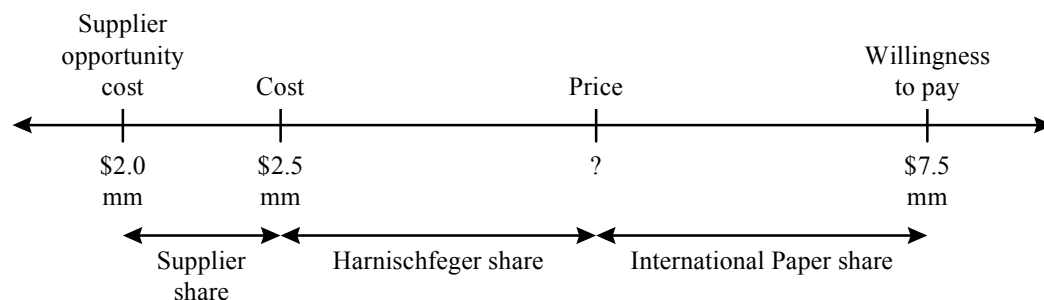
Willingness to Pay and Supplier Opportunity Cost

A customer's *willingness to pay* for a product or service is the maximum amount of money that a customer would be willing to part with in order to obtain the product or service. In the Harnischfeger example, a customer considering the purchase of a portal crane would be willing to pay as much as \$7.5 million for the crane. If it cost more than that, the customer would be better off buying the forklifts for \$1 million and paying the extra operating costs of \$6.5 million.

The concept of *supplier opportunity cost* is precisely symmetrical to willingness to pay. It is the smallest amount that a supplier will accept for the services and resources required to produce a good or service. We call this an "opportunity cost" because it is dictated by the best opportunities that the suppliers have to sell their services and resources elsewhere. In the example, the *actual* cost that Harnischfeger incurred to deliver a portal crane was \$2.5 million. We don't know what the lowest amount the suppliers would have accepted actually was, but we will speculate that it was not far below \$2.5 million, say \$2.0 million.

Imagine that Harnischfeger is bargaining with International Paper, one of the largest paper manufacturers, over the price of a portal crane. For now, suppose that Harnischfeger is the only company that can provide a portal crane and International Paper is the sole customer. The price that emerges from the bargaining may fall anywhere between \$2.5 million, Harnischfeger's cost, and \$7.5 million, International Paper's willingness to pay. (See **Figure 3**.) Our theory says nothing about where the price will fall within this range. If Harnischfeger is a particularly tough bargainer, then the price will tend toward \$7.5 million. If International Paper is the shrewder negotiator, the price will edge toward \$2.5 million.

Figure 3: Division of Value



Source: Brandenburger and Stuart, "Value-Based Business Strategy," 1996

The total *value* created by a transaction is the difference between the customer's willingness to pay and the supplier's opportunity cost. In the example, a sale of a crane to International Paper creates value of \$5.5 million: an item worth \$7.5 million to the customer is created from supplied resources that had a value of only \$2.0 million in their next-best use. The value captured by Harnischfeger is the difference between the negotiated price and \$2.5 million. International Paper captures value equal to \$7.5 million minus the price. And suppliers capture \$0.5 million (**Figure 3**).

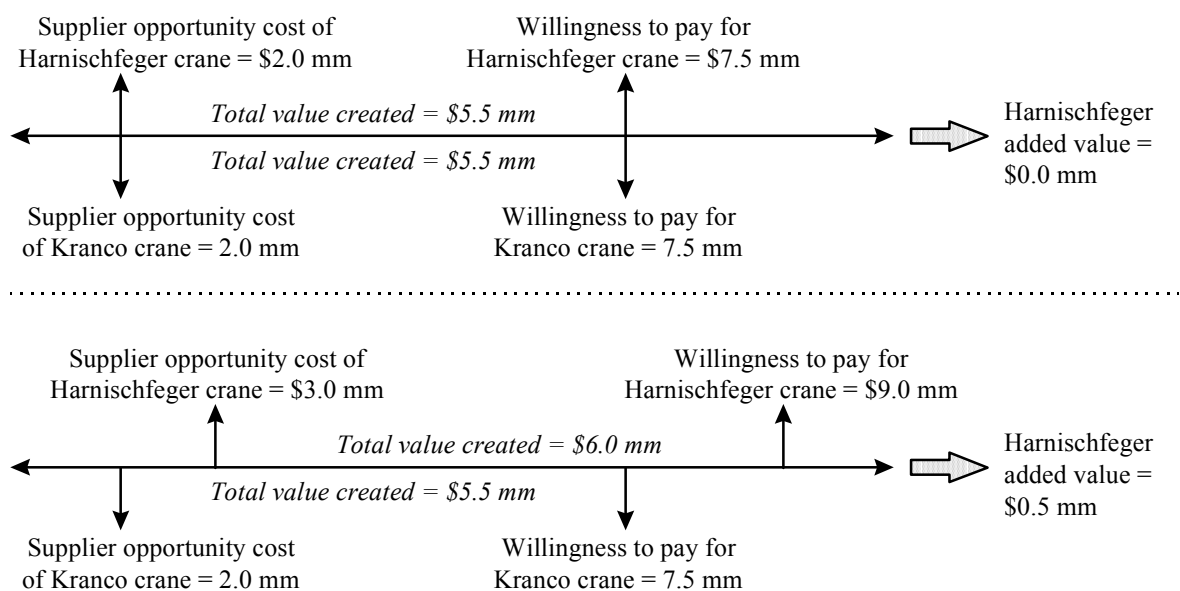
Added Value

A firm's *added value* plays a large role in determining how much value it actually captures. The added value of a firm is the maximal value created by all participants in a transaction minus the maximal value that could be created without the firm. In essence, it is the value that would be lost to the world if the firm disappeared. Consider the situation with Harnischfeger as the sole provider of cranes and International Paper as the only customer. If Harnischfeger opts out of the transaction, the entire \$5.5 million of value goes un-created. Similarly, if International Paper refuses to participate, \$5.5 million of value is no longer generated. Both Harnischfeger and International Paper have an added value of \$5.5 million.

Now consider what happens in the late 1980s when Kranco, a management-buyout firm headed by former Harnischfeger executives, enters the market for portal cranes. Assume that Kranco produces an identical product, with costs of \$2.5 million and supplier opportunity costs of \$2.0 million, and it generates the very same willingness to pay of \$7.5 million. The added value of Harnischfeger is now \$0. If it participates in a deal with International Paper, the total value created is \$5.5 million. If it opts out, Kranco can fill its place, and total value of \$5.5 million is still generated.

Under a condition known as *unrestricted bargaining*, the amount of value a firm can claim cannot exceed its added value. To see why this is so, assume for a moment that a lucky firm *does* strike a deal that allows it to capture more than its added value. Then the value left over for the remaining participants is less than the value that those others could generate by arranging a deal amongst themselves. The remaining participants could break off and form a separate pact that improves their collective lot. Any deal which grants a firm more than its added value is fragile because of such separate pacts. Once Kranco enters, it is not surprising that Harnischfeger captures little value and is barely profitable. After all, it has little or no added value. (See the top half of **Figure 4**.)

Figure 4: Added Value with Harnischfeger and Kranco Providing Cranes



Suppose now that Harnischfeger discovers a way to add some new services to its core product. (See the bottom half of **Figure 4**.) The services boost the willingness to pay of International Paper to \$9.0 million, but, because the services entail additional labor, they raise supplier opportunity costs to \$3.0 million. The total value created with Harnischfeger participating is now \$9.0 million - \$3.0 million = \$6.0 million. The total value if Harnischfeger opts out and Kranco provides the crane is \$7.5 million - \$2.0 million = \$5.5 million. The new service boosts Harnischfeger's added value from \$0 to \$0.5 million, essentially because it raises willingness to pay by more than it increases supplier opportunity costs. By widening the gap between willingness to pay and supplier opportunity cost, Harnischfeger increases the amount of value than it can potentially claim.

The Link to Competitive Advantage

The larger is a firm's added value, the greater is its potential for profit. The logic laid out so far suggests that a firm can boost its added value by widening the wedge it achieves between customer willingness to pay and supplier opportunity cost beyond what rivals attain. We say that a firm with a wider wedge has a competitive advantage in its industry. A firm with a competitive advantage has added value and therefore the potential for profit. The notion of added value highlights the fact that competitive advantage derives fundamentally from *scarcity*. A firm establishes added value by making sure that it is unique in some valuable way—that the network of suppliers, customers, and complementors within which it operates is more productive with it than without it and that it is not readily replaced.

There are two basic ways a firm can establish an advantage. First, the firm can raise customers' willingness to pay for its products without incurring a commensurate increase in supplier opportunity cost. Second, the firm can devise a way to reduce supplier opportunity cost without sacrificing commensurate willingness to pay. Either establishes the wider wedge that defines competitive advantage.

Costs vs. supplier opportunity costs. So far, we have tried to treat buyers, with their willingness to pay, and suppliers, with their opportunity costs, symmetrically. Just as willingness to pay captures the most that buyers will pay for a product, opportunity cost is the least that suppliers will accept for the resources used to make a product. The symmetry is useful: it reminds us that competitive advantage can come from better management of supplier relations, not just from a focus on downstream customers. Recent efforts to streamline supply chains reflect the importance of driving down supplier opportunity costs.

In practice, however, managers often examine *actual* costs, not opportunity costs, because data on actual costs are concrete and available. In the remainder of this note, we focus on the analysis of actual costs. We assume, in essence, that supplier opportunity costs and actual costs track one another closely. A firm's quest for competitive advantage then becomes a search for ways to widen the wedge between actual costs and willingness to pay.

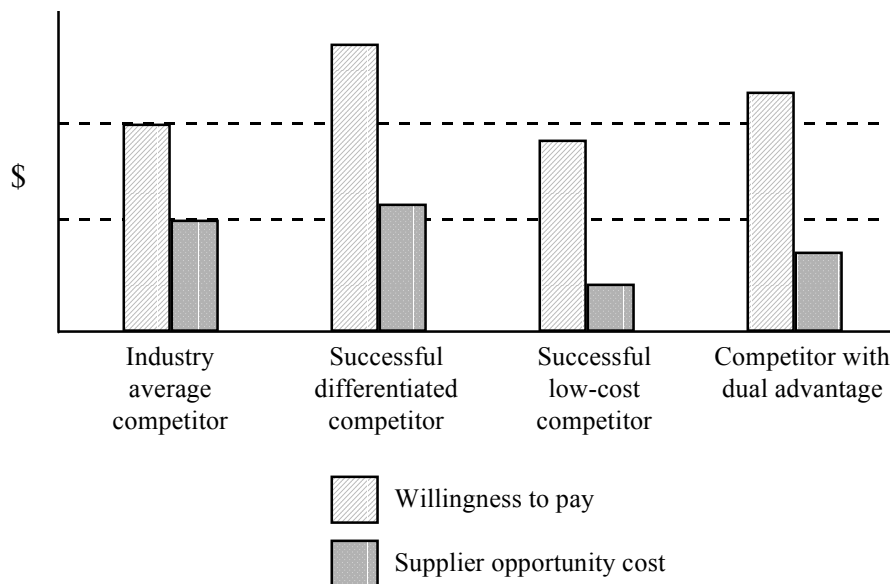
Activity Analysis of Cost and Willingness to Pay¹⁰

The Tension Between Cost and Willingness to Pay

Widening the wedge is difficult because, often, a firm must incur higher costs in order to deliver a product or service for which customers are willing to pay more. Almost all customers would be willing to pay more for a Toyota automobile than for a Hyundai, but the costs of manufacturing a

Toyota are significantly higher than the costs of making a Hyundai. Toyota's higher profit margins derive from the fact that the difference in willingness to pay is greater than the incremental costs associated with its product. As noted above, a firm can achieve a competitive advantage by devising a way to (1) raise willingness to pay a great deal with only slight increases in costs or (2) reap large cost savings with only slight decreases in customer willingness to pay. We call the first a *differentiation strategy* and the second a *low-cost strategy* (Figure 5).¹¹

Figure 5: Types of Competitive Advantage



(The term “differentiated” is often misused. When we say that a firm has differentiated itself, we mean that it has boosted the willingness of customers to pay for its output—that it can command a price premium. We do not mean simply that the company is different from its competitors. Hyundai is certainly different from Toyota, but it is not differentiated with respect to Toyota. Similarly, a common error is to say that a company has differentiated itself by charging a lower price than its rivals. A firm's choice of price does not usually affect how much customers are intrinsically willing to pay for a good.¹²)

The tension between cost and willingness to pay is not absolute: firms can discover ways to produce superior products at lower cost. In the 1970s and 1980s, for instance, Japanese manufacturers in a number of industries found that by reducing defect rates, they could make higher-quality products at lower cost. More recently, Dell has developed a build-to-order model for personal computers that reduces the costs of components, inventories, and obsolescence while also boosting willingness to pay among knowledgeable computer buyers who value the speedy customization that the model permits. Such examples of *dual competitive advantage* are eye-catching and well worth understanding.¹³

Strategy scholars debate, however, how common dual advantages are. Some have argued that dual advantages are rare and are typically based on operational differences across firms that are easily copied.¹⁴ Others contend that breaking the trade-offs between cost and willingness to pay—replacing trade-offs with “trade-ons”—is a fundamental way to transform competition in an

industry.¹⁵ Regardless, it is clear that there is a rich variety of ways to resolve the tension between cost and willingness to pay favorably. Some examples illustrate the possibilities:

- Accenture is regarded as a leader in information-technology consulting because of its deep experience, reinforced by its research and development, its Knowledge Xchange portal, its stringent hiring and training practices, its close relationships with CEOs as well as CIOs of global corporations, and its successful attempts to build its brand. Some of the activities that Accenture undertakes to differentiate itself are clearly costly: R&D and training, for instance, swallow up 5% of revenues, even though they have been cut in recent years. On the other hand, Accenture often faces no competing bids when it pitches consulting work and is able to keep its revenues per consultant high, especially by maintaining a high utilization rate (reportedly 78% in 2003, versus roughly 65% for the industry as a whole).¹⁶ This more than compensates for the extra costs it incurs: Accenture has historically earned returns significantly higher than most other large IT services companies.
- Southwest Airlines has configured itself to focus on budget customers particularly well. It standardizes its fleet around fuel-efficient Boeing 737s, concentrates on short-haul point-to-point routes between midsize cities and secondary airports, offers very low ticket prices and no-frills service (no assigned seats, food service, baggage transfer or connections with other airlines), emphasizes quick turnaround times, and manages to keep its planes in the air one-third longer each day than the average airline. Its stripped-down offering may generate slightly less willingness to pay than the offering of a full-service airline, but it incurs far lower costs than a full-service rival. As a result, Southwest is the only U.S. airline to have been consistently profitable during the last 30 years, has grown at an annual rate of 20–30% over the last five years, and maintains the lowest debt levels among the major carriers.
- Cirque du Soleil is an innovative firm that combines elements of circus and theater. In designing its performances, Cirque excluded many of the high-cost components of traditional circuses—animals, star performers, and three ring shows—and focused on what it considered to be the three elements responsible for the lasting allure of the circus: the clowns, the tent, and the acrobatic acts. By refining the clowns' acts, glamorizing the tent, and incorporating elements from the world of theater—themes and storylines, for example—Cirque de Soleil created a new category of entertainment with which it is synonymous.¹⁷ In essence, the firm stripped out certain costly elements of the traditional circus and added costs in other areas for which a segment of customers is willing to pay a great deal. In 2003, more than 7 million people paid a total of \$650 million to see its live performances, and the value of this privately-held firm was estimated at \$1.2 billion.¹⁸

Activity Analysis

How can one identify opportunities to raise willingness to pay by more than costs or to drive down costs without sacrificing too much willingness to pay? Sheer entrepreneurial insight certainly plays a large role in spotting such opportunities. A Michael Dell sees that customers are becoming comfortable with computer technology, realizes that retail sales channels add more costs than benefits for many customers, and acts on his insight to start a direct-to-the-customer computer business.¹⁹ Or a Liz Claiborne perceives huge pent-up demand for a collection of medium- to high-end work clothes for female professionals.²⁰ Dumb luck also plays a role. Engineers searching for a coating material for missiles in the 1950s discovered the lubricant WD-40, whose sales continued to generate a return on equity between 40% and 50% four decades later.

We believe, however, that smart luck beats dumb luck and analysis can hone insight. To analyze competitive advantage, strategists typically break a firm down into discrete activities or processes and then examine how each contributes to the firm's relative cost position or comparative willingness to pay.²¹ The activities undertaken to design, produce, sell, deliver, and service goods are what ultimately incur costs and generate customer willingness to pay. Differences across firms in activities—differences in what firms actually do day-to-day—produce disparities in cost and willingness to pay and hence dictate competitive advantage. By analyzing a firm activity by activity, managers can (1) understand why the firm does or does not have a competitive advantage, (2) spot opportunities to increase a firm's competitive advantage, and (3) foresee future shifts in competitive advantage.

An analysis of activities usually proceeds in four steps. First, managers of a firm catalog the firm's activities. Second, the managers examine the costs associated with each activity, and they use differences in activities to understand how and why their costs differ from those of competitors. Third, they analyze how each activity generates customer willingness to pay, and they use differences in activities to examine how and why customers are willing to pay more or less for the goods or services of rivals. Finally, the managers consider changes in the firm's activities. The objective is to identify changes that will widen the wedge between costs and willingness to pay. In the following subsections, we discuss these steps in order.

Step 1: Catalog Activities (The Value Chain)

In the remainder of this note, we employ an activity template, the value chain, that can guide managers in breaking down the firm into activities.²² The value chain divides all activities into two classes: primary activities that directly generate a good or service, and support activities that make the primary activities possible. Primary activities are broken down further into inbound logistics, operations, outbound logistics, marketing and sales, and after-sales service. Support activities include procurement of inputs, development of technology and human resources, and general firm infrastructure. **Figure 6** shows the value chain of an Internet start-up that sells compact discs online and ships them by mail to customers.

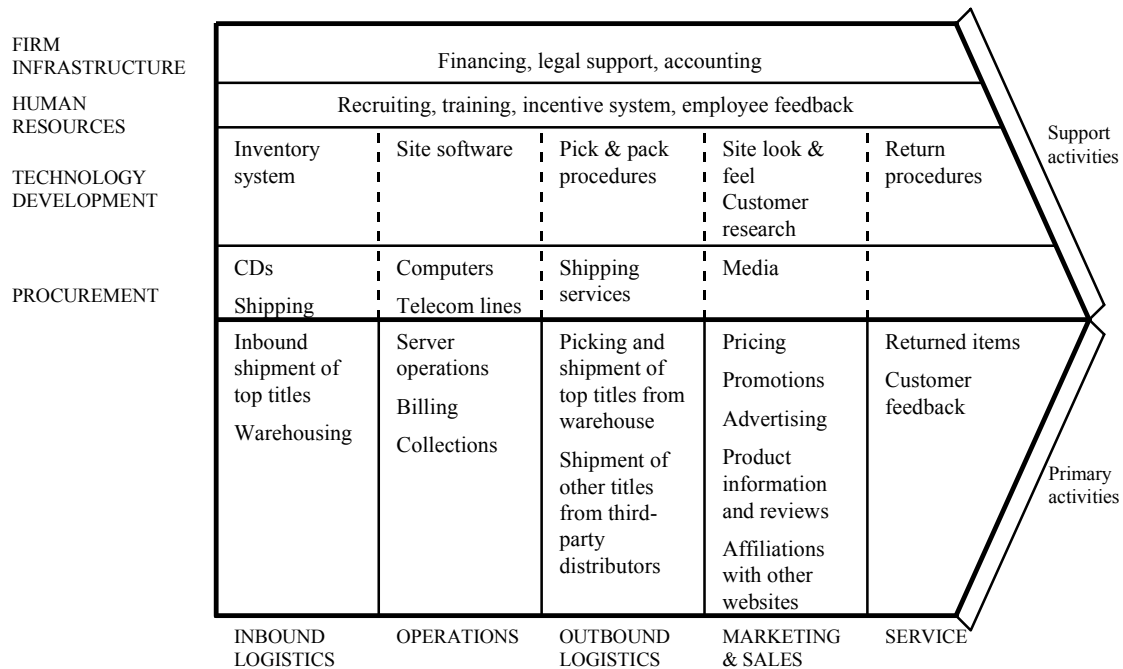
Once activities have been cataloged, they must be analyzed in terms of cost and willingness to pay *relative to the competition*. To illustrate how this is done, we focus on a simple example: the snack cake market in the western region of Canada.* Between 1990 and 1995, Betsy Baking grew its share of this market from a meager 1% to nearly 20%. At the same time, Collins Kitchen, the maker of such long-time favorites as Dinklets and Angel Dogs, saw its dominant 45% share dwindle to 25%. An analysis of relative costs and willingness to pay shows why Betsy Baking and Collins fared so differently.

Step 2: Use Activities to Analyze Relative Costs

Competitive cost analysis is the usual starting point for the strategic analysis of competitive advantage. In pure commodity businesses such as wheat farming, customers refuse to pay a premium for any company's product. In such a setting, a low-cost position is the key to added value and competitive advantage. But even in industries that are not pure commodities, differences in cost often wield a large influence on differences in profitability.

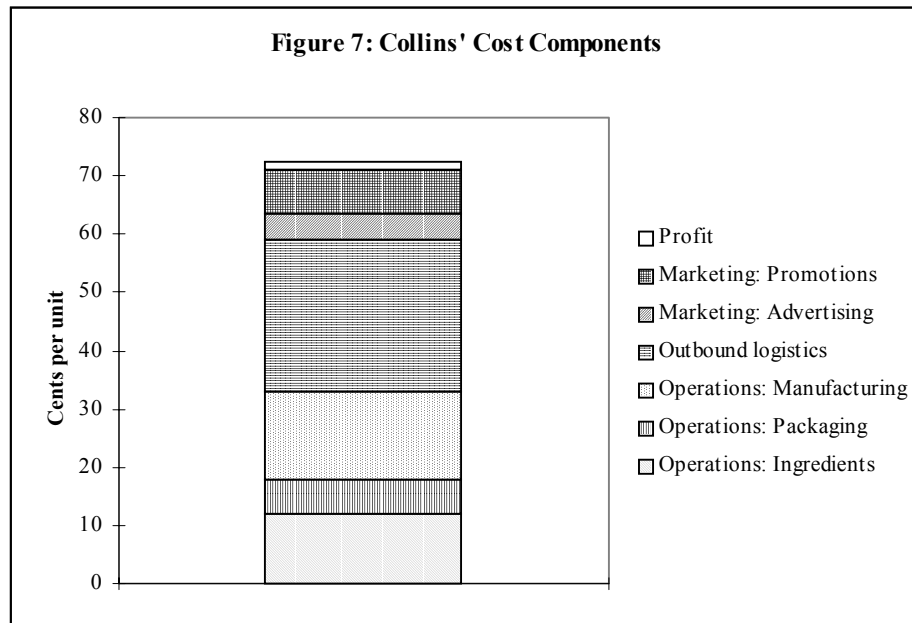
* The authors thank Roger Martin of Monitor Company for this example. Identities of the companies and other items have been altered substantially to protect proprietary information.

Figure 6: Value Chain for an Internet Start-up



Cost analysis was one of the efforts that managers at Collins Kitchens undertook in the mid-1990s as they struggled to understand why their financial performance was poor and their market share plummeting. They cataloged the major elements of their value chain and calculated the costs associated with each class of activities. As **Figure 7** shows, although Collins sold the typical package of snack cakes to retailers for 72¢, raw materials (ingredients and packaging material) accounted for only 18¢ per unit. Operation of automated baking, filling, and packaging production lines, largely depreciation, maintenance, and labor costs, amounted to 15¢. Outbound logistics—delivery of fresh goods directly to convenience stores and supermarkets, and maintenance of shelf space—constituted the largest portion of costs, 26¢. Marketing expenditures on advertising and promotions added another 12¢. A mere penny remained as profits for Collins.

The managers then determined the set of *cost drivers* associated with each activity. Cost drivers are the factors that make the cost of an activity rise or fall. For instance, the managers realized that the cost of outbound logistics per snack cake fell rapidly as a firm increased its local market share; total delivery costs depended largely on the number of stops that a truck driver had to make, and the larger was a firm's market share, the greater was the number of snack cakes a driver could deliver per stop. Urban deliveries tended to be more expensive than suburban because city traffic slowed down drivers. Outbound logistics costs also rose with product variety; a broad product line made it difficult for drivers to restock shelves and remove out-of-date merchandise. Finally, the nature of the product affected logistics costs: snack cakes with more preservatives could be delivered less frequently. The managers developed numerical relationships between activity costs and drivers, for outbound logistics activities and for the other activities in **Figure 7**.

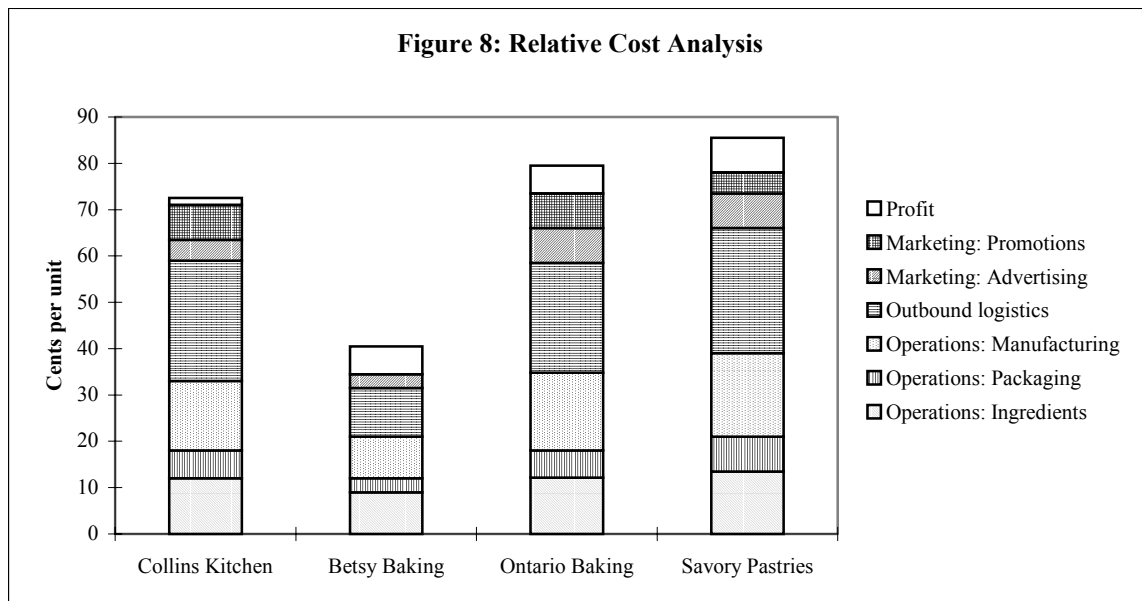


Cost drivers are critical because they allow managers to estimate *competitors'* cost positions. One usually cannot observe a competitor's costs directly, but one can often observe the drivers. One can see, for instance, a competitor's market share, the portion of its sales in urban areas, the breadth of its product line, and the ingredients in its products. Using its own costs and the numerical relationships to cost drivers, a management team can estimate a competitor's cost position.

When Collins' managers did this for Betsy Baking, they found the results sobering. Because Betsy Baking used inexpensive raw material, purchased in bulk, and tapped national scale economies, its operations costs totaled 21¢, in contrast to 33¢ for Collins. Betsy Baking packed its product with preservatives so that deliveries could be made less frequently, kept its product line very simple, and benefited from growing market share. Consequently, its logistics costs per unit were less than half of Collins'. Also, Betsy Baking did not run promotions. Altogether, the managers estimated, a package of Betsy Baking snack cakes cost only 34¢ to produce, deliver, and market. Comparisons with the two other major competitors, Ontario Baking and Savory Pastries, were not so discouraging. Indeed, Collins had a small cost advantage over each. (See **Figure 8**.)

This specific example illustrates a number of general points about relative cost analysis:²³

- When reviewing a relative cost analysis, it is important to focus on differences in individual activities, not just differences in total cost. Ontario Baking and Savory Pastries, for instance, had similar total costs per unit. The two firms had different cost structures, however, and as we will discuss below, these differences reflected distinct competitive positions.
- Good cost analyses typically focus on a subset of all of a firm's activities. The cost analysis in **Figure 8**, for example, did not cover all the activities in the snack cake value chain. Effective cost analyses usually break out in greatest detail and pay the most attention to cost categories that (1) pick up on significant differences across competitors or strategic options, (2) correspond to technically separable activities, or (3) are large enough to influence the overall cost position significantly.



- Activities that account for a thicker slice of costs deserve deeper treatment in terms of cost drivers. For instance, the snack cake managers assigned several cost drivers to outbound logistics and explored these drivers in depth. They spent little time considering the drivers of advertising costs. The analysis of any cost category should focus on the drivers that have the biggest impact on it.
- A particular driver should be modeled only if it is likely to vary across the competitors or the strategic options that will be considered. In the snack cake example, manufacturing location influenced wages rates and therefore operations costs. All of the rivals manufactured their snack cakes in western Canada, however, and manufacturing elsewhere was not an option because shipping was costly and goods had to be delivered quickly. Consequently, manufacturing location was not considered as a cost driver.
- Finally, since the analysis of relative costs inevitably involves a large number of assumptions, sensitivity analysis is crucial. Sensitivity analysis identifies the assumptions that really matter and therefore need to be honed. It also tells the analyst how confident he or she can be in the results. Under any reasonable variation of the assumptions, Betsy Baking had a substantial cost advantage over Collins.

A number of references discuss cost drivers in greater detail and suggest specific ways to model them numerically.²⁴ The catalog of potential drivers is long. Many relate to the size of the firm: economies of scale, economies of experience, economies of scope, capacity utilization, etc. Others relate to differences in firm location, functional policies, timing (e.g., first-mover advantages), institutional factors such as unionization, government regulations such as tariffs, and so forth. Differences in the *resources* possessed by a firm may also drive differences in activity costs. A farm with more productive soil, for instance, will incur lower fertilization costs.

A number of pitfalls commonly snare newcomers to cost analysis. Many companies, particularly ones that produce large numbers of distinct products in a single facility, still have grossly inadequate costing systems that must be cleaned up before they can be used as reference points for estimating competitors' costs. As courses on management accounting point out, conventional accounting

systems often overemphasize manufacturing costs and do a poor job of allocating overhead and other indirect costs. As firms increasingly sell services and transact on the basis of knowledge, these outdated systems make it harder and harder to analyze costs intelligently.²⁵ Also problematic is a tendency to compare costs as a percentage of sales rather than in absolute dollar terms. This confounds cost and price differences. It is also common, but dangerous, to mix together recurring costs and one-time investments. Some analysts confuse differences in firms' costs with differences in their product mixes. One can avoid this problem by comparing the cost positions of comparable products; compare Ford's four-cylinder, mid-sized family sedan to Toyota's four-cylinder, mid-sized family sedan, not some imaginary "average" Ford to some "average" Toyota. Finally, a focus on costs should not crowd out consideration of customer willingness to pay—the topic of the next section.

Step 3: Use Activities to Analyze Relative Willingness to Pay

The activities of a firm do not just generate costs. They also (one hopes) make customers willing to pay for the firm's product or service. Differences in activities account for differences in willingness to pay and hence for competitive advantage and differences in profitability. In general, it appears that differences in willingness to pay account for more of the variation in profitability observed among competitors than do disparities in cost levels.²⁶

Virtually any activity in the value chain can affect customers' willingness to pay for a product.²⁷ Most obviously, the product design and manufacturing activities that influence physical product characteristics—quality, performance, features, aesthetics, durability—affect willingness to pay. Consumers pay a premium for New Balance athletic shoes in part because the firm offers durable shoes in hard-to-find sizes. In fact, by avoiding deals with superstar athletes and publicizing that its shoes are "endorsed by no one," New Balance actively emphasizes to consumers that they should pay attention only to the physical characteristics of its shoes. More subtly, a firm can boost willingness to pay through activities associated with sales or delivery—the ease of purchase, speed of delivery, availability and terms of credit, convenience of the seller, quality of presale advice, etc. In the 1990s, for example, the catalog florist Calyx and Corolla commanded a premium because it delivered flowers faster and fresher than most competitors did.²⁸ Activities associated with post-sale service or complementary goods—customer training, consulting services, spare parts, product warranties, repair service, compatible products—also affect willingness to pay. For example, American consumers may hesitate to buy a Fiat automobile because they fear that spare parts and service will be hard to obtain. Signals conveyed through advertising, packaging, branding efforts, etc. also play a role in determining willingness to pay. Nike's advertising and endorsement activities, for instance, affect the premium it commands. Finally, support activities can have a surprisingly large, if indirect, impact on willingness to pay. The hiring, training, and compensation practices of Nordstrom create a helpful, outgoing sales staff that permits the department store to charge a premium for its clothes.

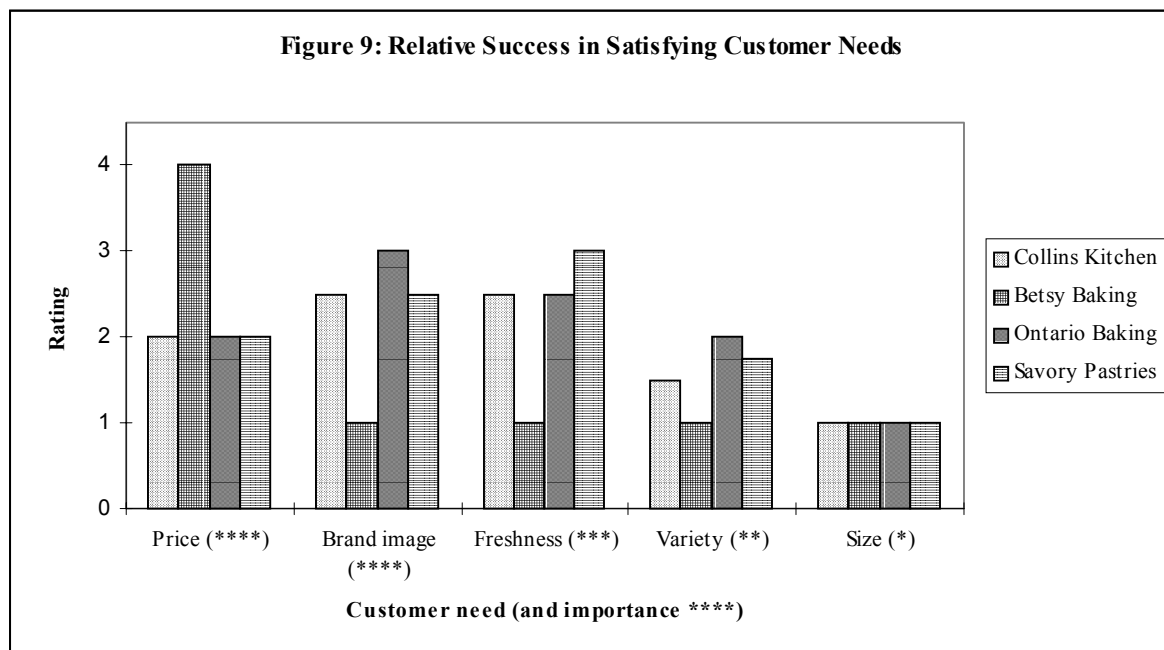
Ideally, a company would like to have a "willingness to pay calculator"—something that tells it how much customers would pay for any combination of activities. For a host of reasons, however, such a calculator is virtually always beyond a firm's grasp. Willingness to pay often depends heavily on intangible factors and perceptions that are hard to measure. Moreover, activities can affect willingness to pay in complicated (i.e., nonlinear and non-additive) ways. And when a business sells to end-users through intermediaries rather than directly, willingness to pay depends on multiple parties.

Lacking a "willingness to pay calculator," most managers who analyze relative willingness to pay do so in a simplified manner. A typical procedure is as follows. First, the managers think carefully about who the *real* buyer is. This can be tricky. In the market for snack cakes, for instance, the

immediate purchaser is a supermarket or convenience store executive. The ultimate consumer is typically a hungry school child. But the pivotal decision maker is probably the parent who chooses among the brands.

Second, the managers work to understand what the buyer or buyers want. The snack cake-buying parent, for example, selects among brands on the basis of price, brand image, freshness, product variety, and the number of servings per box.²⁹ The supermarket or convenience store executive chooses a snack cake on the basis of trade margins, turnover, reliability of delivery, consumer recognition, merchandising support, and so forth. Marketing courses discuss ways to pinpoint such customer needs and desires through formal or informal market research.³⁰ It is important that such research identifies not only what customers *want*, but also what they *are willing to pay for*. Moreover, the research should reveal what the most important needs are and how customers make trade-offs among different needs.

Third, managers assess how successful they and competitors are at fulfilling customer needs. **Figure 9** shows such an analysis for the snack cake market. The analysis helps us understand both the statics and the dynamics of the marketplace. Betsy Baking stands out on an attribute that customers value highly, low price, while Collins is superior on none of the customer needs. This helps us understand the large shifts in market share. Ontario Baking enjoys the best brand image—a position it has paid for via relatively heavy advertising and promotion. (See **Figure 8**.) Savory Pastries delivers the freshest product, reflected in its high manufacturing and raw materials cost. Further analysis, not carried out in the snack cake example, can assign dollar values to the customer needs. For example, it can estimate how much a customer will pay for a product that is one day fresher.



Fourth and finally, the managers relate differences in success in meeting customer needs back to activities. Savory Pastries' high score on the freshness need, for instance, can be tied directly to specific activities regarding procurement and selection of ingredients, manufacturing, and delivery.

At this point, managers should have a refined idea of how activities translate, through customer needs, into willingness to pay. They also understand how activities alter costs. Now they are prepared to take the final step, the analysis of different strategic options. Before we move on to that step, however, we should highlight some guidelines concerning the analysis of willingness to pay.

A major challenge in analyzing willingness to pay is narrowing the long list of customer needs down to a manageable roster. In general, needs that have little effect on customer choice can be ignored. Needs that are equally well satisfied by all current and contemplated products can usually be neglected. If the group of competing products plays a small role in satisfying a need relative to other products outside the group, the need can often be removed from the list.

So far, we have treated all customers as identical. In reality, of course, buyers differ in what they want and how badly they want it. Some customers in a bookstore want novels while others look for business books. (This type of disparity, in which different customers rank products differently, is known as *horizontal differentiation*.) Among those customers who want J.K. Rowling's new *Harry Potter* novel, some are willing to pay for the hardback edition sooner while others will wait for the less expensive soft-cover version. (*Vertical differentiation* arises when customers agree on which product is better—the hardback edition, now—but they differ in how much they will pay for the better product.) The analysis of willingness to pay is trickier, but more interesting, when customers differ in their preferences. The usual response is *segmentation*: one first finds clumps of customers who share preferences and then analyzes willingness to pay segment by segment. In our experience, firms that identify segments pinpoint between two and twelve clusters of customers. The more diverse are customer needs and the cheaper it is to customize the firm's product or service, the more segments a firm typically considers. Some observers have even argued that companies should move beyond segmentation to embrace *mass customization*.³¹ In this approach, enabled by information and production technologies, companies begin to tailor their products to individual customers. Thus, Blinds to Go receives up to 20,000 custom orders for window blinds and shades per day, and it promises to process each order within 48 hours. New approaches to customization have enabled it to build up a business with more than \$100 million in revenues and 20% net margins.³²

Finally, we want to emphasize the limits to analyzing willingness to pay. In some settings, it is possible to quantify willingness to pay quite precisely. For example, when a firm provides an industrial good that saves its customers a well-understood amount of money, it is relatively easy to calculate willingness to pay. (Think of the Harnischfeger example.) Calculations are much more difficult, however, when there is a large subjective component to buyer choice, when customer tastes are evolving rapidly, and when the benefits the customer derives from the product are hard to quantify. A wide range of market research techniques—surveys, hedonic pricing, attribute ratings, conjoint analysis, etc.—are designed to overcome such problems. We remain leary, however, especially when the market research asks people to assess their willingness to pay for new products that they have never seen or for the satisfaction of needs that they themselves may not realize they have. Fine market research “proved” that telephone answering machines would sell poorly, for instance.³³ In some settings, creative insight may have to replace analysis. In all settings, analysis should serve to hone insight, not displace it.

Step 4: Explore Options and Make Choices

The final step in the analysis of cost and willingness to pay is to search for ways to widen the wedge between the two. The management team has the machinery in place to understand how changes in activities will affect competitive advantage. The goal now is to find favorable options. The generation of options is ultimately a creative act, and it is difficult to lay down many guidelines for it. We can, however, suggest a few patterns from past experience:

- It is often helpful to distill the essence of what drives each competitor. Betsy Baking, for instance, saw that preservatives were a substitute for fast delivery. By adding preservatives to its physical product, it could reduce its delivery costs substantially. This also reduced customers' willingness to pay, but the reduction was smaller than the corresponding cost savings for many customers. Such a distillation process often suggests new ways to drive wedges. Savory Pastries, for instance, was tapping a willingness to pay for freshness. The Collins managers, however, felt that Savory was not exploiting this customer need fully; a product even fresher than Savory's might command a large premium, and this might be the basis for a substantial competitive advantage.
- When considering changes in activities, it is crucial to consider competitor reactions. In the snack cake example, the Collins managers felt that Betsy Baking would readily launch a price war against any competitor that tried to match its low-cost, low-price strategy. They were less concerned about an aggressive response from Savory Pastries, whose managers were distracted by an expansion into a different business.
- In crafting alternatives, managers tend to fixate on physical product characteristics and think too narrowly about benefits to buyers. Rarely do they consider the full range of ways in which all of their activities can create a wedge between willingness to pay and costs. One way to avoid a narrow focus is to draw out not only one's own value chain, but also the value chains of one's customers and suppliers and the linkages between the chains.³⁴ Such an exercise can highlight ways to reduce buyers' costs, improve buyers' performance, reduce suppliers' costs, or improve suppliers' performance. Some apparel manufacturers, for instance, have found new ways to satisfy department store buyers, ways that have nothing to do with the physical character of the clothes. By shipping clothes on the proper hangers and in certain containers, the manufacturers can greatly reduce the labor and time required to get clothes from the department store loading dock to the sales floor.
- In rapidly changing markets, it is often valuable to seek options by paying special attention to "bleeding edge" customers—exact customers whose demands presage the needs of the larger marketplace. Yahoo!, the Internet portal, releases test versions of new services to sophisticated users in order to shake down software and sense the future needs of the wider market.³⁵ Similarly, underserved customer segments often point the way to creative alternatives. Circus Circus, the casino operator, built much of its remarkable success in the early 1990s on the insight that Las Vegas offered little to the family-oriented segment of the market. And the Southwest example reminds us that overserved customers can offer an opportunity as well.
- More generally, one of the most potent ways that a firm can alter its wedge between willingness to pay and costs is by adjusting the *scope* of its operations—that is, changing the range of customers it serves or products it offers within an industry. Broad scope in an industry tends to be advantageous when there are significant economies of scale, scope, and learning (including vertical bargaining power based on size), when customers' needs are relatively uniform across market segments, and when it is possible to charge different prices in different segments. Of course, broader isn't always better: there may be diseconomies rather than economies of size, and attempts to serve heterogeneous customers may introduce compromises into a firm's value chain or blur its external or internal message by creating cognitive conflicts in the minds of customers or employees.³⁶ And even when broader *is* better, there tend to be a variety of ways in which a firm can expand its reach, some of which (such as licensing, franchises, or strategic alliances) fall short of an outright expansion of scope.

- Here, we have laid out a process in which a management team develops a comprehensive grasp of how its activities affect costs and willingness to pay, *then* considers options to widen the wedge between the two. In practice, it is often efficient and effective to reverse this process: to start with a set of options, articulate what each option implies for activities, then analyze the impact of each alternative configuration of activities on the wedge between costs and willingness to pay. By reverse-engineering the analyses they do from the options they have, managers can focus on the analyses that truly matter. Of course, this alternative process works best when managers start with a good grasp of the options available to them.³⁷

In general, a firm should scour its value chain for, and eliminate, activities that generate costs without creating commensurate willingness to pay. It should also search for inexpensive ways to generate additional willingness to pay, at least among a segment of customers.

The Whole Versus the Parts

The analysis we have described focuses on decomposing the firm into parts—discrete activities. In the final step of exploring options, however, the management team must work vigilantly to build a vision of the whole. After all, competitive advantage comes from an *integrated set* of choices about activities. A firm whose choices are internally inconsistent is unlikely to succeed.

We have found a *landscape metaphor* helpful to describe the dilemma facing managers who are searching for a favorable set of choices.³⁸ In conceptual terms, the managers of a firm operate in a high-dimensional space of decisions. Each point in this space represents a different set of choices, a different configuration of activities. The elevation corresponding to each point is the added value generated by that configuration. The goal of the senior management team is to guide its firm to a high point on this landscape—a set of decisions that, together, generate a great deal of added value. The search for high ground is made difficult by the fact that the different choices interact with one another: production decisions affect marketing choices, distribution choices need to fit with operations decisions, compensation choices influence a whole range of activities, and so forth. Each interaction implies that a choice made on one dimension affects the cost and willingness-to-pay impact of another choice. Graphically, the interactions make the surface of the landscape rugged with lots of local peaks.

The ruggedness of the landscape has a couple of vital implications. First, it suggests that incremental analysis and incremental change are unlikely to lead a firm to a new, fundamentally higher position. Rather, a firm must usually consider changing many of its activities in unison in order to attain a higher peak. To improve its long-run prospects, a firm may have to step down and tread through a valley. (Consider the wrenching and far-reaching changes required to turn around IBM during the mid-1990s.) Second, the ruggedness implies that there is often more than one internally consistent way to do business within an industry. There is certainly only a limited number of viable positions, but when the interactions among choices are rich, there is usually more than one high peak. In the retail brokerage business, for instance, both Merrill Lynch and Edward Jones succeed, but they do so in very different ways. Merrill Lynch operates large offices in major cities, provides access to a full range of securities, advertises nationally, offers in-house investment vehicles, and serves corporate clients. Edward Jones operates thousands of one-broker offices in rural and suburban areas, handles only conservative securities, markets by means of door-to-door sales calls, produces none of its own investment vehicles, and focuses almost exclusively on individual investors.³⁹ The two firms occupy quite different peaks on the landscape of the financial services industry.

The landscape metaphor reminds us that the creation of competitive advantage involves *choice*. In occupying one peak, a firm foregoes an alternative position. It also highlights the role of competition: it is often more valuable to inhabit one's own, separate peak than to crowd onto a summit that is already heavily populated. Finally, it emphasizes the importance of internal consistency. Peaks are coherent bundles of mutually reinforcing choices.

Conclusion

This note has covered a lot of ground, but the main ideas are fairly simple:

- A successful firm does not simply participate in an attractive industry. It also strives to generate more economic profits than the typical firm in its industry.
- The ability to generate and capture profits in an industry derives from added value. A firm has added value when the network of customers, suppliers, and complementors in which it operates is better off with the firm than without it; the firm offers something that is unique and valuable in the marketplace.
- A firm usually can't claim any value unless it adds some value.
- To have added value, a firm must drive a wedge between customer willingness to pay and supplier opportunity cost—indeed a wider wedge than rivals achieve. A firm that attains a wider wedge is said to have a competitive advantage.
- To establish a competitive advantage, a firm has to do different things than its rivals on a day-to-day basis. These differences in activities, and their effects on relative cost and relative willingness to pay, can be analyzed in detail.
- A firm can use its analysis of activities to generate and assess options for creating competitive advantage. In doing so, the management team must decompose the firm into parts, but also craft a vision of an integrated whole.

Notes

¹ M.E. Porter, *Competitive Strategy*, New York: Free Press, 1980, Chapter 1.

² We are grateful to Uta Werner of Marakon Associates for making available the data for Figures 1 and 2.

³ R. Rumelt, "How Much Does Industry Matter?," *Strategic Management Journal* (12), 1991, pp. 167-185. A.M. McGahan and M.E. Porter, "How Much Does Industry Matter, Really?," *Strategic Management Journal* (18), 1997, summer special issue, pp. 15-30. A.M. McGahan, "The Influence of Competitive Position on Corporate Performance," Harvard Business School mimeo, 1997.

⁴ For further discussion of the definition of competitive advantage, see R. Rumelt, "What in the World is Competitive Advantage," UCLA Working Paper, 2003, and S. Postrel, "Competitive Advantage: A Synthesis," Southern Methodist University Working Paper, 2004.

⁵ A.M. Brandenburger and B.J. Nalebuff, *Co-opetition*, New York: Doubleday, 1996. A.M. Brandenburger and H.W. Stuart, "Value-based Business Strategy," *Journal of Economics and Management Strategy* (5), 1996, pp. 5-24.

⁶ J.L. Heskett, "Scott Cook and Intuit," Harvard Business School Case 396-282, 1996.

⁷ J.W. Rivkin, "Reconcilable Differences: The Relationship Between Industry Conditions and Firm Effects," Harvard Business School Working Paper, 1997.

⁸ This section is based on the work of A.M. Brandenburger and H.W. Stuart, especially the article "Value-Based Business Strategy," *Journal of Economics and Management Strategy* (5), 1996, pp. 5-24. See also M.E. Porter, *Competitive Strategy*, New York: Free Press, 1980, Chapter 2, and A.M. Brandenburger and B.J. Nalebuff, *Co-opetition*, New York: Doubleday, 1996.

⁹ A.M. Brandenburger and H.W. Stuart, "Harnischfeger Industries: Portal Cranes," Harvard Business School Case 391-130, 1992.

¹⁰ This section draws heavily on ideas first developed in M.E. Porter, *Competitive Advantage*, New York: Free Press, especially Chapters 2-4.

¹¹ M.E. Porter, *Competitive Strategy*, New York: Free Press, 1980, Chapter 2.

¹² Exceptions to this rule arise when the price of a good conveys information about it.

¹³ R. Hallowell, "Dual Competitive Advantage in Labor-Dependent Services: Evidence, Analysis, and Implications," in D.E. Bowen, T.A. Swartz, and S.W. Brown (eds.), *Advances in Services Marketing and Management* (6), Greenwich: JAI Press Inc., 1997.

¹⁴ M.E. Porter, *Competitive Strategy*, New York: Free Press, 1980, Chapter 2, and M.E. Porter, "What Is Strategy?," *Harvard Business Review* (74: 6), 1996, pp. 61-78.

¹⁵ A.M. Brandenburger and B.J. Nalebuff, *Co-opetition*, New York: Doubleday, 1996, pp. 127-130.

¹⁶ Accenture Q4 2003 conference call.

¹⁷ W.C. Kim and R. Mauborgne, "Blue Ocean Strategy," *Harvard Business Review* (82: 10), 2004, p. 76.

¹⁸ M. Miller, "The Acrobat," *Forbes*, March 15, 2004, p. 100.

¹⁹ D. Narayandas and V.K. Rangan, "Dell Computer Corporation," Harvard Business School Case 596-058, 1996.

²⁰ N. Siggelkow, "Change in the Presence of Fit: The Rise, the Fall, and the Renaissance of Liz Claiborne," *Academy of Management Journal* (44), 2001, pp. 838-857.

²¹ M.E. Porter, *Competitive Advantage*, New York: Free Press, 1985, Chapters 2-4, and M.E. Porter, "What Is Strategy?," *Harvard Business Review* (74: 6), 1996, pp. 61-78.

²² M.E. Porter, *Competitive Advantage*, New York: Free Press, 1985, Chapter 2.

²³ See P. Ghemawat, *Commitment: The Dynamic of Strategy*, New York: Free Press, 1991, Chapter 4 for a more extensive list of general guidelines.

²⁴ See M.E. Porter, *Competitive Advantage*, New York: Free Press, 1985, Chapter 3 and D. Besanko, D. Dranove, and M. Shanley, *Economics of Strategy*, New York: John Wiley, 1996, Chapter 13.

²⁵ See H.T. Johnson and R.S. Kaplan, *Relevance Lost: The Rise and Fall of Management Accounting*, Boston: Harvard Business School Press, 1987.

²⁶ R.E. Caves and P. Ghemawat, "Identifying Mobility Barriers," *Strategic Management Journal* (13), 1992, pp. 1-12. Of course, this is a general pattern that may or may not hold up in a particular setting.

²⁷ See M.E. Porter, *Competitive Advantage*, New York: Free Press, 1985, Chapter 4 and D. Besanko, D. Dranove, and M. Shanley, *Economics of Strategy*, New York: John Wiley, 1996, Chapter 13.

²⁸ W.J. Salmon and D. Wylie, "Calyx and Corolla," Harvard Business School Case 592-035, 1991.

²⁹ We present "low price" as an attribute that buyers seek. This should not be misunderstood as a statement that price determines willingness to pay. Rather, price is included as an attribute in surveys of customer needs so that one can calibrate the willingness of customers to pay a price premium for the other attributes in the survey (such as freshness).

³⁰ See, for instance, P. Kotler, *Marketing Management: Analysis, Planning, Implementation, and Control*, Englewood Cliffs: Prentice Hall, 1994.

³¹ B.J. Pine, *Mass Customization: The New Frontier in Business Competition*, Boston: Harvard Business School Press, 1993.

³² L. Menor and K. Mark, "Blinds to Go: Invading the Sunshine State," Richard Ivey School of Business Case no. 901D04, 2001.

³³ O. Harari, "The Myths of Market Research," *Small Business Reports*, July, 1994, p. 48 ff.

³⁴ M.E. Porter, *Competitive Advantage*, New York: Free Press, 1985.

³⁵ M. Iansiti and A. MacCormack, "Developing Products on Internet Time," *Harvard Business Review* (75: 5), pp. 108-117.

³⁶ M.E. Porter, "What Is Strategy?," *Harvard Business Review* (74: 6), 1996, pp. 61-78.

³⁷ J.W. Rivkin, "An Options-led Approach to Making Strategic Choices," HBS Note 702-433, 2002.

³⁸ D. Levinthal, "Adaptation on Rugged Landscapes," *Management Science* (43), 1997, pp. 934-950 and J.W. Rivkin, "Imitation of Complex Strategies," *Management Science* (46), 2000, pp. 824-844. The landscape metaphor is derived from evolutionary biology, especially S.A. Kauffman, *The Origins of Order*, Oxford: Oxford University Press, 1993.

³⁹ R. Teitelbaum, "The Wal-Mart of Wall Street," *Fortune*, October 13, 1997, pp. 128-130.